

```

1 import components.simplereader.SimpleReader;
2
3
4
5
6 /**
7  * To calculate the square root.
8  *
9  * @author Eric Zhou.
10 *
11 */
12 public final class Newton4 {
13
14     /**
15      * No argument constructor--private to prevent instantiation.
16      */
17     private Newton4() {
18     }
19
20     /**
21      * Computes estimate of square root of x to within relative error 0.01%.
22      *
23      * @param x
24      *         positive number and zero to compute square root of
25      * @param epInput
26      *         the relevant error and no more than 0.01%
27      * @return estimate of square root
28      */
29     private static double sqrt(double x, double epInput) {
30         final double ep = epInput; //set the error
31         if (x == 0) { //solve the problem that x equals to zero
32             return x;
33         } else {
34             double r = x;
35             while ((Math.abs(r * r - x) / x) > ep * ep) { //calculate the square root
36                 r = (r + (x / r)) / 2;
37             }
38             return r;
39         }
40     }
41
42     /**
43      * Main method.
44      *
45      * @param args
46      *         the command line arguments
47      */
48     public static void main(String[] args) {
49         SimpleReader in = new SimpleReader1l();
50         SimpleWriter out = new SimpleWriter1l();
51         out.println("Enter a eplison: "); // ask for the error
52         double epInput = in.nextDouble();
53         double x = 0;
54         while (x >= 0) { // stop when x is smaller than zero
55             out.println("Enter a non-negative number: ");
56             x = in.nextDouble();
57             out.println(sqrt(x, epInput));
58         }
59
60         /**
61          * Close input and output streams
62          */
63         in.close();
64         out.close();
65     }
66 }
67 }
68

```